

Project Requirement Analysis Phase

Technology Stack (Architecture & Stack)

Date	01 October 2026
Team ID	SWTID-2026-7951
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

The supplied technology-stack template asks for an architectural diagram, application/technology blocks, infrastructure demarcation, external interfaces, data storage, and—if applicable—ML interfaces. It also asks for components/technologies and application characteristics such as security, scalability, availability and performance. [filecite](#)[turn1file5](#)[L8-L19](#)

Technical Architecture

S.No	Component	Description	Technology
1	User Interface	Web interface through which branch users and administrators access records.	ServiceNow Web UI
2	Application Logic-1	Record-level authorization logic that evaluates the current user and record.	ServiceNow ACL
3	Application Logic-2	Conditional access decision based on user branch, record Branch value and administrator	Server-side JavaScript / ACL script

		exception.	
4	Database	Stores application records including the Branch field.	ServiceNow platform database / table
5	User & Role Data	Provides logged-in user identity, roles and branch-related authorization data.	ServiceNow User [sys_user] and role model
6	External API	No external API is required for the described ACL scenario.	Not applicable
7	Machine Learning Model	No ML model is required for field-value access control.	Not applicable
8	Infrastructure	Application and ACL execute within the ServiceNow platform.	ServiceNow instance / cloud platform

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source framework is required; the solution uses the platform ACL mechanism and scripting capability.	ServiceNow ACL / JavaScript

2	Security Implementations	Record-level authorization; branch-value validation; administrator exception; deny-by-default behaviour for users who do not meet the condition.	ACL, roles, server-side script
3	Scalable Architecture	The rule is implemented at the access-control layer and can be reused for additional records or branch conditions.	ServiceNow platform ACL engine
4	Availability	The ACL operates as part of the platform's record access path when the ServiceNow instance is available.	ServiceNow instance
5	Performance	Use a lightweight field/role comparison and avoid unnecessary queries inside the ACL script.	Server-side JavaScript / ACL